neo4j

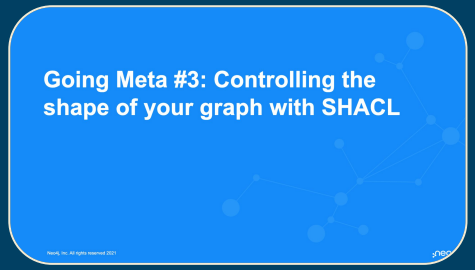
Going Meta

S3 - Episode 6

Validating Neo4j Graphs with SHACL

A popular (and loved) topic of GoingMeta

April 2022



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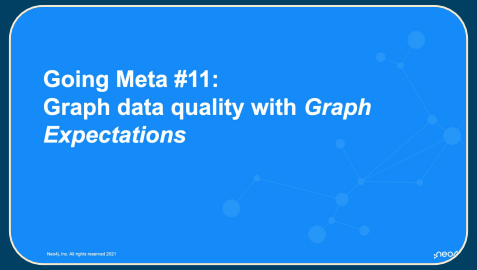

Check them out at:

https://github.com/jbarrasa/goingmeta

What's the problem we are solving?

We want to detect data quality issues (structural or value-related) in a graph without writing custom code.

Examples:

- A customer's phone number must match a regular expression.
- A Product node must be connected to at least three Supplier nodes.
- A Supplier must have exactly one name.
- An Order must not contain properties outside a predefined set.

Refresher: What is SHACL?

SHACL (Shapes Constraint Language) is a W3C recommendation for defining graph constraints.

W3C Recommendation

TABLE OF CONTENTS

1. **Introduction**
1.1 Terminology
1.2 Document Conventions
1.3 Conformance
1.4 SHACL Example
1.5 Relationship between SHACL and RDFS inferencing
1.6 Relationship between SHACL and SPARQL

2. **Shapes and Constraints**
2.1 Shapes
2.1.1 Constraints, Parameters and Constraint Components
2.1.2 Focus Nodes
2.1.3 Targets
2.1.3.1 Node targets (sh:targetNode)
2.1.3.2 Class-based Targets (sh:targetClass)
2.1.3.3 Implicit Class Targets
2.1.3.4 Subjects-of targets (sh:targetSubjectsOf)
2.1.3.5 Objects-of targets (sh:targetObjectsOf)
2.1.4 Declaring the Severity of a Shape
2.1.5 Declaring Messages for a Shape
2.1.6 Deactivating a Shape
2.2 Node Shapes
2.3 Property Shapes
2.3.1 SHACL Property Paths
2.3.1.1 Predicate Paths
2.3.1.2 Sequence Paths
2.3.1.3 Alternative Paths

Shapes Constraint Language (SHACL)

W3C Recommendation 20 July 2017

This version:
https://www.w3.org/TR/2017/REC-shacl-20170720/

Latest published version:
https://www.w3.org/TR/shacl/

Latest editor's draft:
https://w3c.github.io/data-shapes/shacl/

Implementation report:
https://w3c.github.io/data-shapes/data-shapes-test-suite/

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Test Suite:
SHACL Test Suite

Please check the **errata** for any errors or issues reported since publication.

See also **translations**.

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https://www.w3.org/TR/shacl/

What does it look like?

```
@prefix ex: <http://example.neo4j.com/graphvalidation#> .
@prefix sh: <http://www.w3.org/ns/shacl#> .
@prefix neo4j: <neo4j://graph.schema#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
```

Namespaces are important in RDF

```
ex:ProdShape a sh:NodeShape ;
  sh:targetClass neo4j:Product ;
```

We are defining the shape of nodes labelled as **Product**

```
sh:property [
  sh:path neo4j:productName ;
  sh:pattern "^\\w[\\s\\w\\.]*$" ;
  sh:maxCount 1 ;
  sh:datatype xsd:string ;
];
```

The property **productName** is optional (max card but no min), is of type string and must match this regular expression...

```
sh:property [
  sh:path neo4j:unitPrice ;
  sh:minExclusive 10 ;
  sh:maxInclusive 100 ;
  sh:maxCount 1 ;
];
```

...

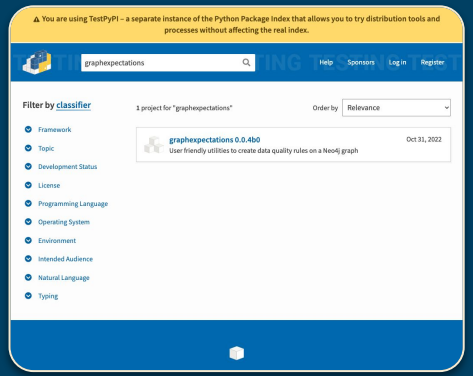
```
sh:property [
  sh:path neo4j:supplied_by ;
  sh:class neo4j:Supplier ;
  sh:minCount 2 ;
];
```

The relationship **supplied_by** has a minimum cardinality of 2 and must point to a node of type Supplier

You may not want to write SHACL…

You can use the graphexpectations package

https://test.pypi.org/search/?q=graphexpectations



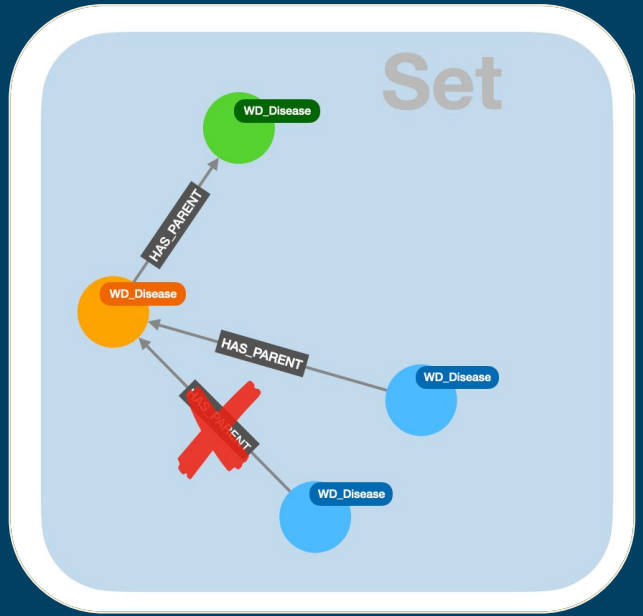

```
import graphexpectations as ge
import pandas as pd
```

```
companyExpectations = ge.Set(nodeType="Company")
companyExpectations.expect_property_values_to_be_of_type(property="foundingDate", datatype="date")
companyExpectations.expect_number_of_property_values_to_be_between(property="foundingDate", min=1,
companyExpectations.expect_outgoing_relationship_to_connect_to_nodes_of_type(relationship="location
companyExpectations.expect_node_types_to_be_in_list(typeList=["Company"])

personExpectations = ge.Set(nodeType="Person")
personExpectations.expect_node_types_to_be_in_list(typeList=["Person"])
```

```
s = ge.Suite(desc="suite of expectations for my Neo4j Company
s.add_expectations([personExpectations, companyExpectations, l
```

Sets: describing a the shape of a set of nodes



Associated to a node type (Label)

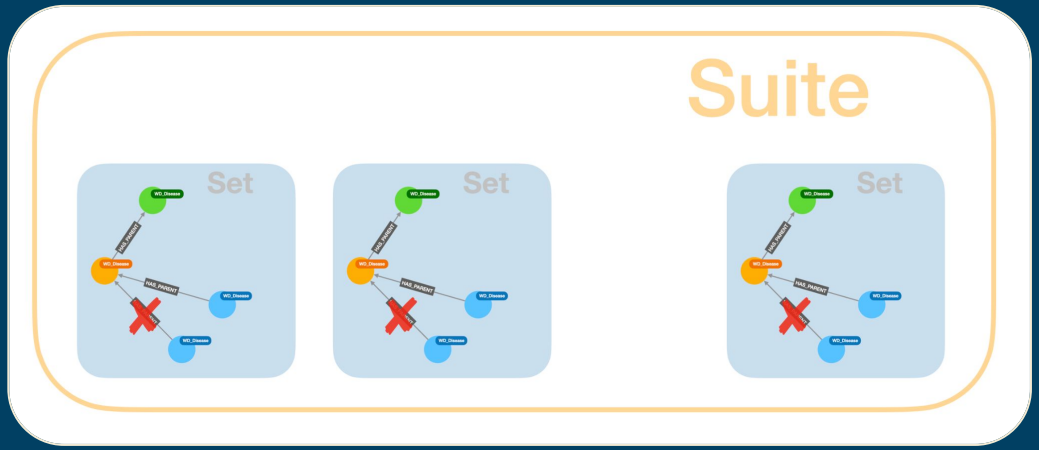
```
companyExpectations = ge.Set(nodeType="Company")
```

Or to a set of nodes defined by an expression (query)

```
locationExpectations = ge.Set(query=
        " (focus)<-[:location|locationCity]-() ")
```

(equivalent to a sh:NodeShape in SHACL)

Suites: a collection of Sets



Create and add sets...

```
s = ge.Suite(desc="suite of expectations for my Neo4j Company KG")
s.add_expectations([personExpectations, companyExpectations, locationExpectations])
```

Let’s try it!